JENNIFER ZHU

MEDICAL IMAGING & COMPUTATIONAL RESEARCH ANALYST

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PROFILE

UCSF-trained biomedical researcher with an M.S. in Health Data Science and B.S. in Statistics. Hands-on experience in longitudinal quantitative MRI, radiopathomic feature analysis, image quality control, and clinical outcome modeling. Seeking dry-lab computational imaging and research roles.

RESEARCH EXPERIENCE

Graduate Student Researcher | UCSF, Lupo Lab | June 2025–Present

- Analyze longitudinal multiparametric MRI and clinical outcomes across 95 patients with newly diagnosed glioblastoma from three research cohorts, covering pre- and post-radiotherapy, pre-progression, and progression scans.
- Design and adapt ROI-based quantitative imaging feature workflows for contrast-enhancing and non-enhancing lesions, including distributional statistics, tumor-volume changes, and radiopathomic cellularity and tumor-probability maps.
- Quantify threshold-defined abnormal regions using intensity thresholds, morphological cleaning, connected-component filtering, volumetric features, and imaging quality control.
- Perform Kaplan–Meier, multivariable Cox regression, linear regression, model diagnostics, and false discovery rate adjustment to assess imaging–outcome associations.
- Create MRI overlays, survival plots, manuscript figures, and scientific summaries for multidisciplinary collaborators; first-author manuscript in preparation.

SELECTED PROJECTS

I-SPY2 — Bayesian Modeling of Treatment Response

- Fit Bayesian logistic regression models of pathological complete response using baseline MRI tumor diameter, sphericity and HR/HER2 status.
- Compared three models using leave-one-out cross-validation; assessed posterior distributions, MCMC convergence and prior sensitivity on 384 complete cases.

BloodMNIST — Biomedical Computer Vision

- Collaborative project comparing a PyTorch convolutional neural network with a random-forest baseline on eight classes of microscopy images.
- CNN achieved 90.76% held-out test accuracy; evaluated precision, recall, F1 and ROC-AUC across classes.

Breast Tumor Classification — PCA & QDA

- Applied PCA and quadratic discriminant analysis to Wisconsin breast cancer cell-nucleus features.
- Reported 93.9% test accuracy and 0.991 ROC-AUC; assessed class-specific recall and false-negative risk.

EDUCATION

M.S., Health Data Science | University of California, San Francisco | June 2026

B.S., Statistics | San Francisco State University | December 2022

TECHNICAL SKILLS

Imaging: Quantitative MRI; ROI/region-level feature extraction; NIfTI processing; imaging QC; segmentation; longitudinal MRI; radiopathomic cellularity/tumor probability maps

Statistics: Cox regression; Kaplan–Meier; linear regression; multivariable adjustment; FDR control; longitudinal comparisons

Programming & Visualization: Python; R; SQL; MATLAB; pandas; NumPy; Matplotlib; ggplot2; Jupyter; R Markdown; Git; Linux

RESEARCH OUTPUT

First-author manuscript in preparation — longitudinal radiopathomic MRI features and clinical outcomes in glioblastoma (UCSF Lupo Lab).